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disruption or infiltration of systems by rival states, either directly or through proxy organizations. As famously exemplified by the massive 2007 attack against Estonia's Internet infrastructure—which took down online financial services, media, and government agencies—the more advanced and Internet-based an economy, the more devastating cyberattacks can be. Estonia is among the most digitalized societies in the world (see “E-stonia Takes Off” in the March 2018 *F&D*).

Countermeasures

If critical infrastructure—say, a power grid—or telecommunication and transportation networks are affected, or an attack prevents governments from collecting taxes or providing critical services, major disruptions with systemic economic implications could ensue and potentially pose a public health or security hazard. In such instances, the aggregate risk to the global economy could exceed the sum of individuals' risks, because of the global nature of IT networks and platforms, the national nature of response structures, ineffective international cooperation, or even the presence of nation-states among the attackers.

International cooperation in combating and prosecuting cybercrime lags well behind the global nature of the threat. The best way to tackle cybercrime is to attack its business model, which relies on the exceptional risk/reward ratio associated with ineffective prosecution. In this context, the business risk of cybercrime must be raised significantly, but this is possible only with better international cooperation.

Cybercrime operations can span several jurisdictions, which makes them harder to take down and prosecute. Some jurisdictions are slow, ineffective, or simply uncooperative in tackling cybercrime. Stronger cooperation would make tracking down suspects and charging them faster and more effective.

In the financial sector, regulators have developed specific assessment standards, set enforceable

expectations and benchmarks, and encouraged information sharing and collaboration among firms and regulators. Bank regulators conduct IT examinations that factor cybersecurity preparedness into stress testing, resolution planning, and safety and soundness supervision. Some require simulated cyberattacks designed specifically for each firm, drawing on government and private sector intelligence and expertise, to determine resilience against an attack. Companies have also increased investment in cybersecurity and are incorporating cybersecurity preparedness into risk management. In addition, some have sought to transfer some risk via cyber insurance.

The current cybersecurity landscape remains disparate and decentralized, with risks handled mainly as local idiosyncratic problems. There are some cooperation mechanisms, and governments and regulators are stepping up their efforts, but the choice of cybersecurity is largely determined by corporate need—“each to its own.” This must change to bring about generally enhanced cyber risk resilience. Strong preventive measures are needed both at the regulatory and technology levels and across industries. Among the most important of these is adherence to minimum cybersecurity standards, enforced in a coordinated way by regulators. Stepped-up cybersecurity awareness training will help defend against the basic technical weaknesses and user errors that are the source of most breaches.

Cyberattacks and cybersecurity breaches seem inevitable, so we also need to focus on how fast we detect breaches, how effectively we respond, and how soon we get operations back on track. **FD**

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What Are Cryptocurrencies?

A potential new form of money offers benefits while posing risks

Antoine Bouveret and Vikram Haksar

HUNDREDS OF THEM have sprouted, with fanciful names like Primecoin, Dash, and Verge. They have developed cult-like followings among the tech-savvy. Their values fluctuate wildly. Some people say these mysterious bits of computer code will someday replace money as we know it. What exactly are these cryptocurrencies, and what makes people think they are worth anything at all? To answer these questions, let's first look at how money evolved.

Uses of money

Money serves as a store of value, a means of exchange for goods and services, and a unit of account that measures value. Before money, human societies exchanged goods and services directly—a bushel of grain for a pig, say. This was not very efficient. As societies grew more complex, commodity monies were developed—from sea-shells to copper, silver, and gold. Some states introduced fiat money—which has no intrinsic value other than the promise to pay—such as paper money in eighth century China under the Tang dynasty.

Most early forms of fiat money were neither very stable nor widely accepted, as people did not believe the issuer would honor its commitment to redeem the money. Governments were tempted to print more money to buy goods or raise wages, which fueled inflation (think of people moving cash around in wheelbarrows in post–World War I Germany). Modern central banks seek to maintain price stability by regulating the supply of money on behalf of governments.

Bookkeeping and ledgers

An increasingly extensive and complex financial system gave rise to the need for trusted intermediaries and credible accounting systems. The



development of double-entry bookkeeping in Renaissance Italy was a major innovation that strengthened the role of large private banks. In modern times, central banks emerged at the apex of payment systems. With computerized bank ledgers, the coordinating role of central banks increased.

How do such ledgers work? Financial institutions adjust the positions of their account holders in their internal ledgers, while the central bank validates transactions among financial institutions in a central ledger. For example, Mehrnaz uses money from her account in bank A to buy goods from Mary, who has an account in bank B. Bank A debits the money from Mehrnaz's account. The central bank moves money from bank A to bank B and records the transaction in its central ledger. Bank B then adds the money to Mary's account. As you can see, the system is based on trust in the central bank and in its ability to safeguard the integrity of the central ledger and ensure that the same money is not spent twice.